The range is the _____ set of a function.

Select the correct option

○	input
○	output



Click to Save An

MTH532 – Complex Analysis and Differential Geometry (Quiz No. 3)

Quiz Start Time: 08:45 PM, 27 May 2024

Question # 1 of 10 (Start time: 08:45:42 PM, 27 May 2024)

Total Marks:

The limit $\lim_{z \to 0} \frac{z}{\bar{z}}$ _____ .

Select the correct option

Reload Math Equations

exists

does not exist

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 03:10:40 PM, 27 May 2024)

The function $f(z) = z^2$ is called _____ function.

Select the correct option

○	square root
○	square



Click

Question # 2 of 10 (Start time: 07:51:19 PM, 27 May 2024) Total Marks:

Each branch of hyperbola

$$x^2 - y^2 = c_1 \qquad (c_1 > 0)$$

is mapped in one-to-one manner onto _____.

Select the correct option

Reload Math Equations

- the vertical line $u = c_1$
- the horizontal line $v = c_1$

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 07:58:26 PM, 26 May 2024)

The function $u(x,y) = \frac{xy}{x^2+y^2}$ does not have a limit when _____.

Select the correct option

◉	$(x,y) \to (0,0)$
○	$(x,y) \to (0,1)$
○	$(x,y) \to (1,0)$
○	$(x,y) \to (1,1)$

uestion # 10 of 10 (Start time: 06:12:00 PM, 26 May 2024)

Domain of the function

$$f(z) = \frac{1}{z^2 - 1}$$

is _____.

lect the correct option

- $z \neq \pm 2$
- $z \neq 0$
- $z \neq \pm 1$ ✓
- $z \neq \pm i$

10 (Start time: 06:11:10 PM, 26 May 2024)

$$\text{The value of } w = \frac{1}{z} \text{ when } z = 2 - i \text{ is}$$

ct option

$\frac{1}{5}(2 + i)$



$\frac{1}{5}(2 - i)$

Question # 7 of 10 (Start time: 07:46:08 PM, 27 May 2024)

If $f(z) = z^2$, then $f(x + iy) =$ _____.

Select the correct option

Reloc

- $x^2 - y^2 + 2xyi$
- $x^2 - y^2 i - 2xyi$
- $x^2 + y^2 i - 2xyi$
- $x^2 + y^2 - 2xyi$

Click to Save Answer & Move

Question # 1 of 10 (Start time: 07:50:05 PM, 27 May 2024) Total Marks:

Mapping is the _____ way of visualizing a given function.

Select the correct option

Reload Math Equations

- textual
- geometrical
- analytical
- numerical

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 07:51:13 PM, 27 May 2024)

For which values of z is $w = \frac{1}{z}$ defined?

Select the correct option

Reloa

○	w is defined for all $z = 0$.
○	w is defined for all $z \neq 0$.

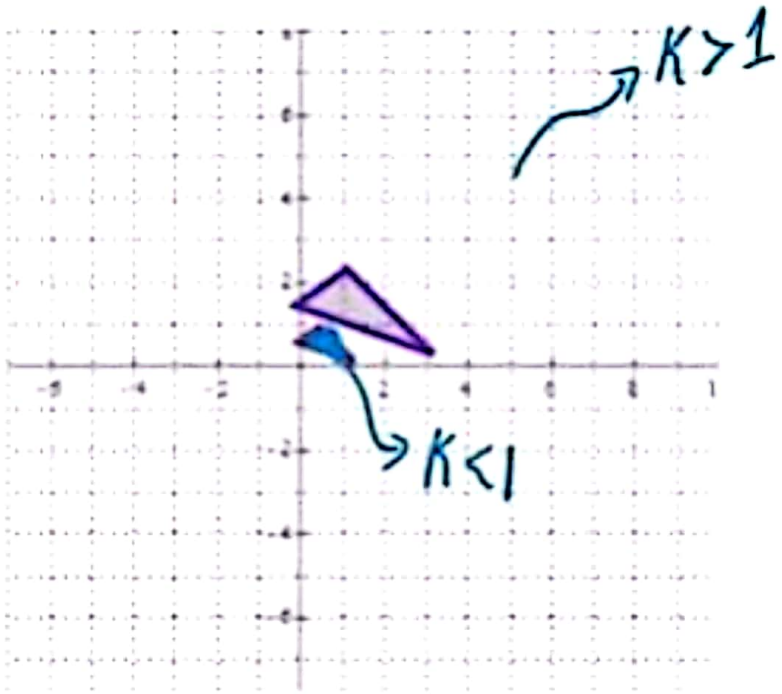
Click to Save Answer & Move t

It has following two cases:

Case 1: If $k < l$, it reduces the distance between the points by factor k.

Case 2: If $k > 1$, it stretches the distance between the points by factor k.

125



15.44

Remark

Domain of polynomial function is the whole complex plane.

LINEAR POLYNOMIAL FUNCTION

$$f(z) = az + b$$

where a, b are complex numbers, is called linear polynomial function, or just linear transformation

Example

$f(z) = 1 + z + z^2$ is a polynomial function.

Example

$f(z) = 1 + z + z^2 + z^3 + z^4 + \ldots$ is not a polynomial. It is an infinite series

Example

$f(z) = \frac{1}{z} = z^{-1}$ is not a polynomial.

RATIONAL FUNCTIONS

A rational function is an expression of the following form

$$\frac{P(z)}{Q(z)}$$

where $P(z)$ and $Q(z)$ are polynomials.

LINEAR FRACTIONAL TRANSFORMATIONS

A linear fractional transformation is of the form

$$T(z) = \frac{az + b}{cz + d} \quad , \quad ad - bc \neq 0$$

where $a, b, c, d \in C$

Question # 1 of 10 (Start time: 02:59:34 PM, 27 May 2024) Total Marks: 1

In the transformation $w = kz$, if $k < 1$, then it _____the distance between the points by factor k.

Select the correct option

Reload Math Equations

○	stretches
○	reduces



Click to Save Answer & Move to Next Question

Domain of a polynomial function is the whole complex plane.

elect the correct option

Reload Mat

- True
- False

Click to Save Answer & Move to Next

The function $f(z) = 1 + z + z^2$ is_____

lect the correct option

- not a polynomial.
- a polynomial. ✓

Question # 5 of 10 (Start time: 07:59:08 PM, 27 May 2024) **Total M**

When we apply the function $f(z) = z^2$ on the lines parallel to imaginary or real ax
then the images will be _____ in w-plane.

Select the correct option

Reload Math Equati

- (●) parabola
- ( ) hyperbola

Click to Save Answer & Move to Next Questi

Question # 10 of 10 (Start time: 07:58:53 PM, 27 May 2024)

Total Marks

If f approaches two complex numbers $L_1 \neq L_2$ for two different curves or paths through z_0 then the $\lim_{z \to z_0} f(z)$ _____

Select the correct option

Reload Math Equations

- exists
- does not exist



Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 07:59:41 PM, 27 May 2024)

The domain of the function $f(z) = \frac{1}{(z-2)^4}$ is _____.

Select the correct option

Reload

○	$C \setminus \{2\}$
○	$C \setminus \{3\}$
○	$C \setminus \{1\}$
○	$C \setminus \{0\}$

Click to Save Answer & Move to Ne

Question # 8 of 10 (Start time: 07:56:04 PM, 27 May 2024)

$$\lim_{(x,y)\to(1,2)} 3xy^2 = _____ .$$

Select the correct option

Reload

○	3
○	6
○	9
○	12

Click to Save Answer & Move to

In the complex valued function $w = f(z)$, the dependent variable is _____.

lect the correct option

- w
- z

Click to Save Answ

Consider $\lim_{z \to a} f(z) = A$ and $\lim_{z \to a} g(z) = B$. Then $\lim_{z \to a} (f(z) \pm g(z)) =$ _____.

he correct option

Re

$A + B$

$A \pm B$

A

$A - B$

Click to Save Answer & Ne

Question # 10 of 10 (Start time: 07:57:08 PM, 27 May 2024) Tot

The real and imaginary parts of the polar form of the complex function $f(z) = z^4$ are _____.

Select the correct option

Reload Math Eq

○	$U(r, \theta) = r^2 \cos 4\theta,\ V(r, \theta) = r^2 \sin 4\theta$
○	$U(r, \theta) = r^2 \cos 3\theta,\ V(r, \theta) = r^2 \sin 3\theta$
○	$U(r, \theta) = r^4 \cos 3\theta,\ V(r, \theta) = r^4 \sin 3\theta$
○	$U(r, \theta) = r^4 \cos 4\theta,\ V(r, \theta) = r^4 \sin 4\theta$

Click to Save Answer & Move to Next Qu

estion # 8 of 10 (start time: 08:00:32 PM, 27 May 2024)

The $\frac{z}{z}$ does not exist as z approaches 0.

ect the correct option

Reloc

True

False

Click to Save Answer & Move to

The domain is the _____ set of a function.

elect the correct option

- output
- input

Click to Save Answer &

The linear transformation is a composition of _____

elect the correct option

Reload

○	rotation and magnification
○	translation and magnification
○	translation and rotation
○	translation, rotation and magnification ✓

Click to Save Answer & Move to

If $f(z) = z^2$ then $f(x+iy) =$

$x^2 - y^2 + 2xyi$

The domain is the input set of the function

The range is the output set of the function

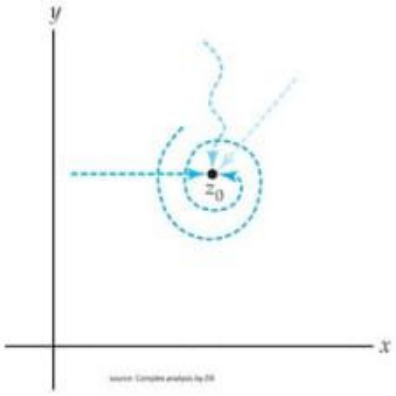
Which of the following are important part of a function domain and value

The domain of the function $f(z) = \frac{1}{2z - \bar{z}}$ is $\mathbb{C} \setminus \{$

4
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21:50 Vo LTE Vo LTE 337 B/s

If f approaches two complex numbers L_1 and L_2, where $L_1 \neq L_2$, for two different curves or paths through z_0, then $\lim_{z \to z_0} f(z)$ does not exist.



EXAMPLE 1

Show that $\lim_{z \to 0} \frac{z}{\bar{z}}$ does not exist.

SOLUTION

First we calculate the limit when z approaches 0 along the real axis, put $y = 0$

$$\lim_{z \to 0} \frac{z}{\bar{z}} = \lim_{x \to 0} \frac{x + i0}{x - i0} = \lim_{x \to 0} \frac{x}{x} = \lim_{x \to 0} 1 = 1$$

If z approaches 0 along the imaginary axis, then put $x = 0$

$$\lim_{z \to 0} \frac{z}{\bar{z}} = \lim_{y \to 0} \frac{0 + iy}{0 - iy} = \lim_{y \to 0} \frac{iy}{-iy} = \lim_{y \to 0} -1 = -1$$

Both limits are not equal, so the limit does not exist.

Remark

The above criteria is used to show non-existence of limit.

Remark

But for existence of limits, we cannot use the above criteria to show existence of same limit on two different paths.

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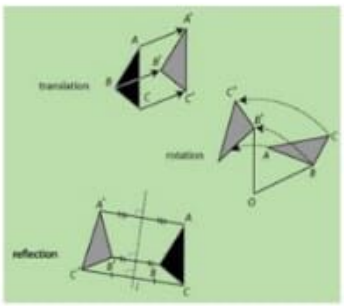
21:47 Vo LTE Vo LTE 316 B/s

LECTURE No. 10

MODULE No. 20

MAPPINGS LINEAR TRANSFORMATION

PART I



So far we have discussed, what is a complex valued function, and what is a mapping associated to it?

A mapping is a geometrical way of visualizing a given complex valued function. It helps in understanding the behavior of a function.

Now we are going to discuss a particular type of functions and the mapping associated to them.

First example is of linear transformation which is very simple but important complex valued function. They have very interesting geometrical interpretations as well.

LINEAR TRANSFORMATION

Linear function is the linear polynomial. It is a complex valued function of the form

$$L(z) = Az + B$$

where A, B are fixed complex numbers.

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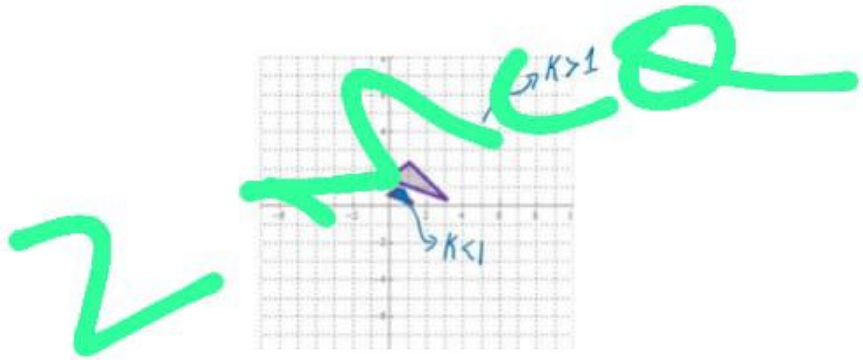
15:46 Vo LTE Vo LTE 3.8 K/s

It has following two cases:

Case 1: If $k < l$, it reduces the distance between the points by factor$k$.

Case 2: If $k > 1$, it stretches the distance between the points by factor$k$.

125



In other words, if $k = 1$, then there is no magnification.

if $k < 1$, it makes the size of object smaller.

If $k > 1$, it makes the size of object bigger.

GENERALIZED CONCEPT

We have the linear transformation

$$\omega = L(Z) = Az + B$$

If $A = ke^{i\alpha}$ and $B = a + ib$, where $k > o$ is a positive real number, then

$$L(z) = ke^{i\alpha}z + B$$

Here k is responsible for magnification. α is responsible for rotation, B is responsible for translatio

In terms of Composition of a rotation, a magnification and a translation, it can be written as

the effect of rotating the plane through an angle by $\alpha = Arg\, A$.

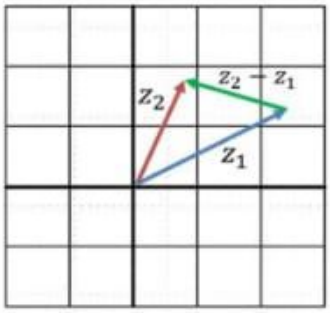
followed by a magnification by the factor $k = |A|$

and followed by a translation by the vector a by $B = a + ib$.

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and is a real number.

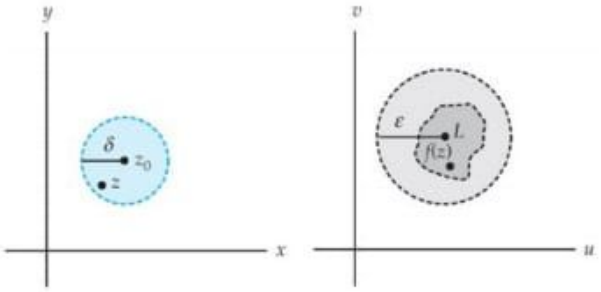


The limit of f exists and is equal to L as $z \to z_0$ if

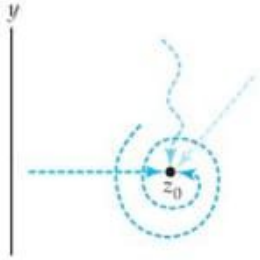
"For every $\varepsilon > 0$, there exists $\delta > 0$ such $|f(z) - L| < \varepsilon$ whenever $0 < |x - x_0| < \delta$".

This fact is denoted by

$$\lim_{z \to z_0} f(z) = L$$



There are many ways to approach z_0 in complex plane. We can choose any path the neighbourhood of z_0 to reach z_0.



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21:47

image in $\omega -$ plane.

Now we are going to study some other types of complex valued functions such $\boldsymbol{\omega = z^n}$ and $\boldsymbol{\omega = z^{\frac{1}{n}}}$. These are the special types of functions which are called power functions. We have already studied the power function in Calculus as a real valued function.

A real valued function of the form $\boldsymbol{f(x) = x^a}$ where a is a real constant, is power function.

Complex Power Function

A complex power function is a function of the form

$$\boldsymbol{f(z) = z^{\alpha}}$$

where $\boldsymbol{\alpha}$ is a complex constant if we consider $\boldsymbol{\alpha}$ where $\boldsymbol{i}$ is involved, then the situation becomes complicated. Such cases we will discuss later.

At the moment, we focus our attention at $\boldsymbol{\alpha}$ as an +ve integer. Here we choose $\boldsymbol{\alpha = 2}$

$$\boldsymbol{\omega = f(z) = z^2} \quad \text{(Square function)}$$

$$\boldsymbol{\omega = f(z) = z^{\frac{1}{2}}} \quad \text{(Square Root Function)}$$

We will discuss different properties and geometry of these functions. For simplicity, we will use exponential form of complex function.

The Mapping $\boldsymbol{f(z) = z^2}$

Given function is

$$\omega = f(z) = z^2$$

Put $z = re^{i\theta}$, we get

$$\omega = (re^{i\theta})^2$$

$$\omega = r^2 e^{i2\theta}$$

Let $\omega = \zeta e^{i\theta}$, then $f(z)$ is given by the following equations

$$\rho = r^2, \quad \varphi = 2\theta$$

For $r > 1$, the bigger circle in $\omega -$ plane

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15:44 Vo LTE Vo LTE 2.3 K/s
50

Remark

Domain of polynomial function is the whole complex plane.

LINEAR POLYNOMIAL FUNCTION

$$f(z) = az + b$$

where a, b are complex numbers, is called linear polynomial function, or just linear transformation.

Example

$f(z) = 1 + z + z^2$ is a polynomial function.

Example

$f(z) = 1 + z + z^2 + z^3 + z^4 + \ldots$ is not a polynomial. It is an infinite series

Example

$f(z) = \frac{1}{z} = z^{-1}$ is not a polynomial.

RATIONAL FUNCTIONS

A rational function is an expression of the following form

$$\frac{P(z)}{Q(z)}$$

where $P(z)$ and $Q(z)$ are polynomials.

LINEAR FRACTIONAL TRANSFORMATIONS

A linear fractional transformation is of the form

$$T(z) = \frac{az + b}{cz + d} \quad , \quad ad - bc \neq 0$$

where $a, b, c, d \in C$

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15:52 Vo LTE Vo LTE 1.2 K/s

In the previous module, we saw different properties and geometry of linear transformation, where we took a subset A in the $z-$ plane, then we apply complex valued function on A and we see its effect and image in $\omega-$ plane.

Now we are going to study some other types of complex valued functions such $\omega = z^n$ and $\omega = z^{\frac{1}{n}}$. These are the special types of functions which are called power functions. We have already studied the power function in Calculus as a real valued function.

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$$\omega = f(z) = z^2$$

Put $z = re^{i\theta}$, we get

$$\omega = (re^{i\theta})^2$$

$$\omega = r^2 e^{i2\theta}$$

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21:47 Vo LTE Vo LTE 12.1 K/s

If $A = ke^{i\alpha}$ and $B = a + ib$, where $k > o$ is a positive real number, then

$$L(z) = ke^{i\alpha}z + B$$

Here k is responsible for magnification, α is responsible for rotation, B is responsible for translation.

In terms of Composition of a rotation, a magnification and a translation, it can be written as

the effect of rotating the plane through an angle by $\alpha = Arg\ A$.

followed by a magnification by the factor $k = |A|$

and followed by a translation by the vector a by $B = a + ib$.

REMARK

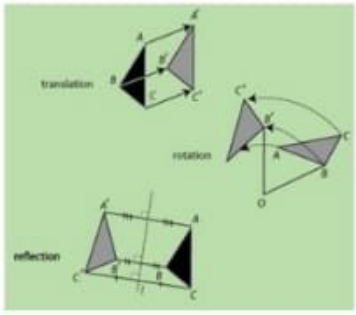
The linear transformation is composition of

126

(1) Translation

(2)Rotation

(3) Magnification



INVERSE TRANSFORMATION

L is a one-to-one mapping from the z-Plane onto ω − plane, so its inverse transformation $L^{-1}(z)$ is possible.

$\because \omega = L(z) = Az + B \qquad \because \omega = L(z) \Rightarrow L^{-1}(\omega) = z$

$\omega = Az + B$

$\omega - B = Az$

$\frac{\omega}{A} - \frac{B}{A} = z$

$z = L^{-1}(\omega) = \frac{\omega}{A} - \frac{B}{A}$

$L^{-1}(\omega) = \frac{\omega}{A} - \frac{B}{A}$

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15:44

where the a_i are comple... mber...

Remark

Domain of polynomial function is the whole complex plane.

LINEAR POLYNOMIAL FUNCTION

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LINEAR FRACTIONAL TRANSFORMATIONS

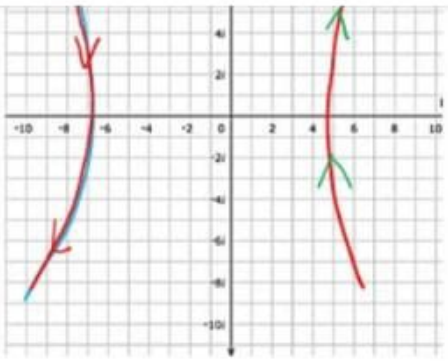
A linear fractional transformation is of the form

$$T(z) = \frac{az + b}{cz + d}, \quad ad - bc \neq 0$$

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22:06 Vo LTE Vo LTE 830 B/s

transformed





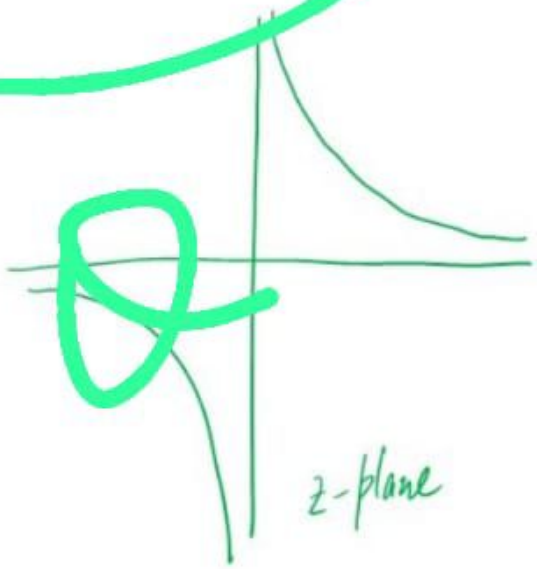
The parametric representation for the image of the left branch of the hyperbola expr

$$u = c_1,\ v = -2y\sqrt{c_1 + y^2} \qquad (-\infty < y < \infty)$$

Ea branch of hyperbola

$$2xy = c_2 \qquad (c_2 > 0)$$

i transformed into the line horizontal line $v = c_2$



From $v = 2xy$ we get $\boldsymbol{v = c_2}$

By using $\boldsymbol{y = \frac{c_2}{2x}}$ in $u = x^2 - y^2$, we get

149

$$u = x^2 - \left(\frac{c_2}{2x}\right)^2 = x^2 - \frac{c_2^2}{4x^2}$$

$$v = c_2 \qquad 0 < x < \infty$$